

Comparing Transformer Neural Architecture to Recurrent Neural Networks for Natural Language Understanding

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Abstract- This research report explores the comparative efficacy of Recurrent Neural Networks (RNNs) and Transformer Neural Architectures in Natural Language Understanding (NLU). Focusing on word embedding vectors and model architecture, the study dissects their impact on language understanding models. A thorough comparison highlights the strengths and weaknesses of RNNs and Transformers, considering factors like training efficiency and scalability. An illustrative example exposes limitations in Long Short-Term Memory (LSTM) networks, emphasizing the need for alternative architectures. Real-world use cases underscore the practical implications, aiding researchers, and practitioners in selecting optimal models for diverse NLU applications.

Index Terms- Long Short-Term Memory, Natural Language Understanding, Recurrent Neural Network, Transformer Neural Architecture.

I. INTRODUCTION

The transformer neural network is a novel neural architecture that excels in handling sequential data by employing self-attention mechanisms, enabling parallelization and efficient learning of complex relationships in input sequences. It was first proposed in “Attention Is All You Need” published in 2017 and has since then been a popular technique for Natural Language Understanding (NLU) Tasks such as text summarization, sentiment analysis, language translation and paraphrase detection. Through its intricate neural architecture consisting of decode block structure and multi-head attention system, this neural network is able to outperform the vanilla recurrent neural networks. RNNs (Recurrent Neural Networks) are feed-forward neural networks. So, the output of the previous step is the input of the current step. In natural language understanding tasks such as sentiment analysis, where the order and sequence of words matter the process of using data from the previous step aids in determining the sentiment of the text. This research report will delve into the neural architecture of Vanilla recurrent neural networks, comparing them with transformer neural network architecture, to ultimately decide which model is more suitable for natural language understanding tasks.

II. VANILLA RECURRENT NEURAL NETWORKS

Vanilla Recurrent Neural Networks (RNNs) are a type of neural network architecture designed for sequential data processing, making them particularly relevant for natural language understanding tasks. Unlike traditional feedforward networks, RNNs have connections that form cycles, allowing them to capture temporal dependencies in sequential data. However, vanilla RNNs suffer from challenges such as vanishing or exploding gradients, limiting their ability to effectively capture long-term dependencies in language sequences.

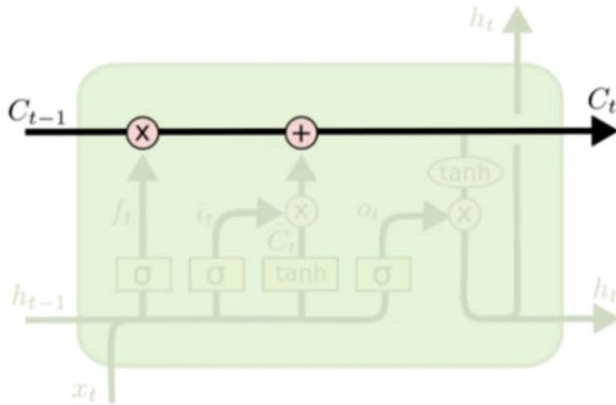
For computer machines to make sense of language, they need to covert words into vectors before feeding them into a model. These vectors are known as word embeddings. These word embeddings are processed in certain ways by models to perform specific tasks such as sentiment analysis, or text summarization. Recurrent neural networks came into existence because making sense of a sentence requires contextual knowledge, or in other words information about previous words. The repeated cells in a Vanilla RNN receive an input word embedding along with the hidden state (H_t) from the past sequence. Using this, depending on the task, it can be used to display informed outputs.

For Natural Language Understanding, a specific type of model called the sequence-to-sequence model is used. Sequence-to-sequence RNNs enable the transformation of input sequences into output sequences, crucial for tasks like machine translation. They consist of an encoder that processes the input sequence and a decoder that generates the output sequence. This architecture has proven effective for various natural language processing tasks as it provides a framework for handling variable-length input and output sequences.

Overall, the main benefit of recurrent neural networks in natural language understanding was that it could account for previous word embeddings when making choices. However, feature means that each steps prediction depends on the previous output, hence causing difficulties in parallelizing it's operations. Additionally, long sequences, or sentences, can result in errors due to the vanishing gradient or yield explosion. In other words, the memory is not capable enough to remember older word embeddings. These issues are addressed by Long Short-Term Memory (LSTM) Networks.

III. LONG SHORT-TERM MEMORY NETWORKS

The LSTM Network is a type of recurrent neural network (RNN) designed specifically to overcome the issue of vanishing gradients, which typically result in slow training of data. With this model, storing information for longer periods is a default and aids in learning long-term dependencies. A single LSTM neuron can be represented through the diagram below:



SOURCE: <https://colah.github.io/posts/2015-08-UNDERSTANDING-LSTMs/>

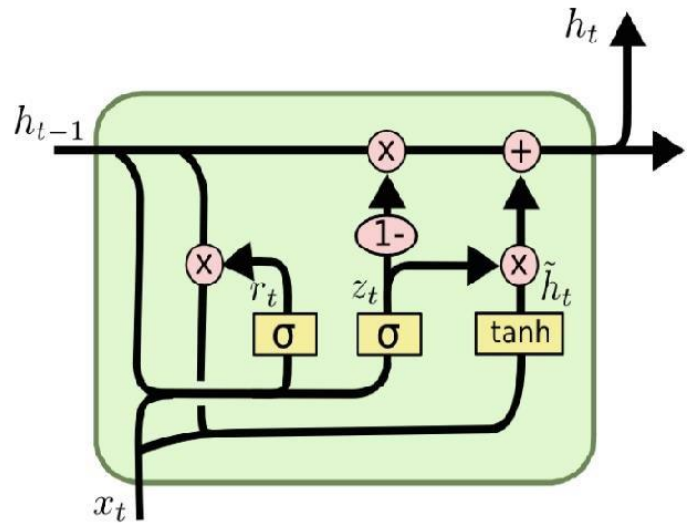
The main difference between vanilla RNNs and LSTM networks is the branch that runs from C_{t-1} to C_t . This is called the cell state. It carried the information to solve the “long-term dependencies” problem, by avoiding the long processing between four interacting layers.

IV. GATED RECURRENT UNIT (GRU)

GRU offers similar performance to LSTM, but with a simpler architecture. Compared to Gated recurrent units, LSTMs have more parameters.

GRU has a single hidden state that fulfills both functions, which might restrict its capacity, whereas LSTM has distinct cell states and outputs, allowing it to store and output different information. Additionally, there's a chance that LSTM and GRU will react differently to the hyperparameters, like learning rate, dropout rate, and sequence length. When comparing the number of gates used by both models, LSTM models use three gates including input, forget, and output to manage information flow, while GRU models use two gates: update and reset, which help improve efficiency. GRU models, while less powerful, may be better suited in certain situations. Since they have lower training times and perform faster, they can be implemented in tasks which involve real-time application such as speech recognition. Additionally, in scenarios where memory is limited, GRU models may be preferred. Overfitting is a common problem faced to lack of sufficient data that the model can be trained with. It is less common in GRU models as they have a relatively less complex structure.

A Diagram of a GRU model is shown below:



SOURCE: https://www.researchgate.net/figure/A-typical-GRU-Block-Diagram_fig1_339016510

V. ATTENTION

The concept of attention allows machines to focus on specific parts of the input to make meaning of the text. This results in a more precise output while making the process more efficient. The sequence-to-sequence model was restricted to using the “vector” at the very last encoder RNN cell, which was then interpreted by the decoder. However, some parts present in intermediary cells of the RNN may be more useful and efficient in finding the final output sequence. Here, “Attention is all you need”. A seq-2-seq model with attention, using their hidden states, each decoder token will examine every encoder token to determine which ones require additional attention. Moreover, a classic seq-2-seq model transferred far less data from encoder to decoder, whereas a seq-2-seq model with the main advantage of Attention is that it accounts for the appropriate encoded representations regardless of the position of input tokens. However, to do this, an extra computation step of adding weights to each encoded representation is needed.

Self-attention, specific kind of attention, focuses on ability of finding context from each token in a set compared to all other tokens in the set. A token here is what represents the vector embedding assigned to a word. First, the model represents each word in the sequence as a vector. This allows simple words to be represented in a more complex three-dimensional structure allowing a more detailed picture of its meaning. Then, it creates three additional vectors for each word: query, key, and value. To compare with other vectors, it calculates a “similarity score” between the query of one word and the key of others, and then uses these scores to create a weighted sum of the value vectors, essentially attending to the most relevant words for understanding the current one. This process repeats itself, eventually giving the machine a deeper understanding of the string provided, and thus being a useful tool for natural language understanding.

The fundamental technique for comprehending relationships within sequences is self-attention. This is expanded upon by multi-head attention, which adds complex layers that result in richer representations and, in some cases, improved performance. In comparison, multi-head typically uses 4-8 heads, each performing the action of self-attention, with its own three vectors of query, key and value. This allows the model to focus on various interdependencies such as part-of-speech information or syntactic features. Due to the numerous heads, there will be many outputs. These are concentrated and further processed to formulate a more precise and detailed representation of the sentence.

Both these mechanisms serve as the basis of the Transformer Neural Architecture in natural language understanding.

VI. TRANSFORMER NEURAL ARCHITECTURE.

Transformer neural networks have emerged as a breakthrough in natural language processing (NLP), addressing the limitations posed by recurrent neural networks (RNNs) through innovative techniques such as attention mechanisms. Unlike RNNs, Transformers operate in parallel, allowing for more efficient processing of input sequences. The introduction of attention mechanisms in Transformers enables them to selectively focus on relevant parts of the input, capturing long-range dependencies more effectively. This attention-based approach not only overcomes the vanishing gradient problem but also facilitates parallelization, making Transformers highly scalable. Consequently, Transformers have become the cornerstone of state-of-the-art NLP models, achieving remarkable performance in tasks such as machine translation, text summarization, and language understanding. The versatility and effectiveness of Transformers have propelled them to the forefront of NLP research and applications, revolutionizing how we process and understand human language.

VII. CONCLUSION

This study has compared Transformer neural architectures with Recurrent Neural Networks (RNNs) for Natural Language Understanding (NLU). The findings highlight the superior performance of Transformers in capturing long-range dependencies and improving contextual understanding in NLU tasks. While Transformers offer scalability and parallelization advantages, the choice between architectures should consider factors like dataset size and task requirements. This research underscores the need for ongoing exploration in neural architecture design to enhance NLU capabilities, suggesting potential for hybrid models or architectural refinements to further advance the field.

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